

Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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- **Economic Problem:** Selecting best options under costly evaluation
 - Determining each option's true quality costs decision-maker time and effort
 - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
 - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
 - Graduate programs invest time and money into students' development
 - If programs are undervaluing applicants with certain characteristics, future such applicants may receive currently-withheld opportunities

Research Questions

- ① Are admissions committees optimizing, or are they making mistakes?
 - Goal is to accept applicants with highest success probabilities who will matriculate
 - Per literature, measure success with program completion and job placement ranking
 - Predictors of acceptance only (success only) may be overweighted (underweighted)
 - ② If they are making mistakes, how can they make better admissions decisions?
 - In turn, by changing decision rules, how much better of a cohort can they admit?
- ★ Methods apply to undergraduate/professional admissions, hiring, and other settings

Methodology and Main Results

- Build structural models of admissions committee's decision problem that support:
 - Statistical tests of whether committee's decision rule aligns with desired objective
 - Counterfactual simulations quantifying gains from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
 - For most applicants, estimated success probabilities align with those implied by committee's decisions. But some applicant characteristics are misweighted
 - Correctly weighting these characteristics would increase admitted cohort's average success probability by about 4 percentage points without reading more files
 - Estimate corrected for winner's curse $\Rightarrow$ reflects out-of-sample performance

Contribution to Literature

- **Graduate Admissions:** Regress acceptance and success on applicant characteristics
 - Krueger & Wu (2000); Athey, Katz, Krueger, Levitt & Poterba (2007); Stock, Finegan & Siegfried (2006–2015); Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
 - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
 - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)
 - **Optimality:** Cho & Rust (2010); Hoffman, Kahn & Li (2018); Kleinberg, Lakkaraju, Leskovec, Ludwig & Mullainathan (2018); Anderson, Rosen, Rust & Wong (2025)
- ★ **This Paper:** Builds and estimates dynamic selection models of admissions, which test optimality by relating acceptance effects to success effects

Synthetic Data Protocol (IRB-Approved)

① **Administrative office receives data from university departments**

- Algorithmically extracts variables from textual files, such as recommendation letters

② **Synthetic data (Patki et al. 2016) used for model development**

- Estimate joint distribution of variables, and simulate synthetic dataset from it
- Rows do not correspond to real subjects, preventing identity disclosure
- Cells with < 5 comparable observations binned to prevent attribute disclosure

③ **Administrative office executes final estimation code on actual data**

- Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

Outline

- 1 Regression Analysis
- 2 Structural Models
- 3 Structural Estimation
- 4 Optimality and Gains

Data Description

- **Sample:** Applicants to research university's economics PhD program from 2015–2019
- **Outcomes:** Measures of admission and success
 - **Admission:** Read File $\Rightarrow$ Accept $\Rightarrow$ Matriculate $\Rightarrow$ Success
 - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Covariates:** Objective vs. subjective variables
 - **Objective:** Demographics and performance measures, such as GRE scores
 - **Subjective:** Textual processing of recommendation letters, statements of purpose, CVs, and supplemental forms using dictionaries to construct variables

Regression Results

- **Method:** Logistic regressions for acceptance, matriculation, and success
 - **Success:** *Top 100 Placement* = 1 for 20% of applicants vs. 34% of matriculants
- **Results:** Different independent variables best predict each dependent variable
 - **Accept:** *Committee Score* is best predictor. Other predictors: demographics, *GRE Quantitative Percentile*, *Work Experience*, *CV Coding* (negative)
 - **Matriculate:** Predictors of acceptance often negatively predict matriculation. Best predictor of matriculation is *Open House* attendance
 - **Success:** *PhD Program Rank* is best predictor. Other predictors: *Age*, *TOEFL/IELTS*, *Work Experience*, *Essay Positive Terms (%)*

Acceptance vs. Success AMEs

Dependent Variable	Accept	T100 Placement
Winsorized Age	-0.0001	0.0053*
Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

» Insufficiency of Comparing AMEs

» Diversity Objective

» Regressions vs. Model

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to test optimality
- Need model of admissions to relate acceptance effects to success effects

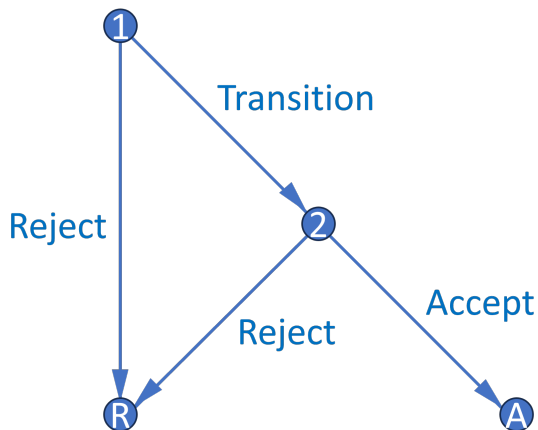
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- 1 Regression Analysis
- 2 Structural Models**
- 3 Structural Estimation
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Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
 - Per literature, committee should accept applicants with highest success probabilities
 - Committees also accept applicants with higher matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
 - Quantify gains from deviating from actual to counterfactual decision rule

Dynamics



- **Round 1:** See only objective variables. Transition to Round 2 or reject
- **Round 2:** See survivors' objective and subjective variables. Accept or reject
- Assume for now number of applicants transitioned and accepted is fixed
- Matriculation model will account for success and matriculation probabilities

Portfolio Choice Problem

- Accept applicants with highest success probabilities under capacity constraints
- For applicant n , choose $d_{1,n} \in \{\text{Transition}, \text{Reject}\}$ and $d_{2,n} \in \{\text{Accept}, \text{Reject}\}$:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N_T} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_A,$$

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N \underbrace{EV_{2,n}(x_n; \bar{\lambda}_2^*)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

where x = objective variables, z = subjective variables, $y \in \{\text{Success}, \text{Failure}\}$,
 N = number of applicants, (N_T, N_A) = max number of files read and acceptances

Portfolio Choice Problem

$$\begin{aligned} & \max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N_T} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_A \\ & \max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N \underbrace{EV_{2,n}(x_n; \bar{\lambda}_2^*)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T, \end{aligned}$$

- Portfolio choice problem has cutoff rule solution
 - **Round 2:** Accept transitioned applicant if success probability exceeds Round 2 cutoff
 - **Round 1:** Transition applicant if Round 2 expected value exceeds Round 1 cutoff
 - Decisions across applicants independent, so no need to consider all subsets

Cutoff Rule Model

- ★ **Goal:** Use choices and outcomes data to estimate restricted success probabilities, while respecting constraint on number of files committee can read
- ★ **Legend:** Estimated model objects vs. Choices and outcomes data
- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Cutoff Rule Model

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Cutoff Rule Model

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Cutoff Rule Model

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

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Conditional Choice Probabilities:

$$P(d_1 | x, t, \bar{k}) = \frac{\exp(u_1(\cdot, d_1)/\sigma)}{\sum_{\tilde{d}_1 \in \{T, R\}} \exp(u_1(\cdot, \tilde{d}_1)/\sigma)}, \quad P(d_2 | x, z, t, \bar{k}) = \frac{\exp(u_2(\cdot, d_2)/\sigma)}{\sum_{\tilde{d}_2 \in \{A, R\}} \exp(u_2(\cdot, \tilde{d}_2)/\sigma)}$$

►► Dynamic Parameterization

►► Static Model

Portfolio Choice with Matriculation

- Accept applicants with highest success probabilities who are expected to matriculate
- For applicant $n \in \{1, \dots, N\}$, choose $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^N \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M,$$

where (x, z) = variables, $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$,
 N = number of applicants, N_M = max number of matriculants

Portfolio Choice with Matriculation

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

such that: $\sum_{n=1}^N \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M$

Theorem 1

*For large N , this problem has a cutoff rule solution on **success** probabilities.*

- Likely matriculants more valuable, but also more expensive $\Rightarrow$ effects cancel
- Matriculation probabilities matter in Round 1 of a two-round model

Cutoff Rule with Matriculation

- ★ **Goal:** Two-round cutoff rule model where matriculation matters
- ★ **Legend:** Estimated model objects vs. Choices and outcomes data
- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = cutoffs

Cutoff Rule with Matriculation

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = cutoffs

Cutoff Rule with Matriculation

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S, m_1^* = M | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = cutoffs

Cutoff Rule with Matriculation

Round 1:

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Importance of Matriculation:

$$\lim_{\sigma \rightarrow 0} EV_2 = \mathbb{E}_{\tilde{z}}[\max\{P(y = S | x, \tilde{z}, t, \bar{k}) - P(y_2^* = S | t), 0\} P(m = M | x, \tilde{z}, t) | x, t, \bar{k}]$$

- Matriculation matters when expected success probability $\gg$ Round 2 cutoff
- **Upshot:** Low-ranked programs must consider matriculation in Round 1

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- 3 Structural Estimation**
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Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^N \underbrace{f(z_n | x_n, t_n; \theta_F)}_{\text{Density of } z|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$
$$\prod_{n=1}^N \underbrace{P_{d_{1,n}}(\theta_T)^{\mathbb{1}(d_{1,n}=T)} (1 - P_{d_{1,n}}(\theta_T))^{\mathbb{1}(d_{1,n}=R)}}_{\text{Round 1 conditional choice probabilities}} \prod_{n=1}^{N_T} \underbrace{P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Round 2 conditional choice probabilities}}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with flexible logistic specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- $\mathcal{L}^R$ aligns success probabilities implied by outcomes with those implied by CCPs

Cutoff Probability Identification

- Identify each year's cutoffs by how selective committee is that year
 - More acceptances and stronger applicants increase cutoff
 - Fewer acceptances and weaker applicants decrease cutoff
- While selectivity identifies cutoffs, sorting improves restricted likelihood

Theorem 2

Cutoff probability estimate is such that number of acceptances N_A equals expected N_A . For any cutoff and N_A , likelihood is highest when committee accepts best N_A applicants.

Restricted Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.3731 (0.3067, 0.4539)	36.58% (29.81%, 43.92%)	
⋮	⋮	⋮	
2019	0.3887 (0.3268, 0.4624)	38.14% (31.89%, 44.81%)	
Sigma			0.0020 (0.0013)

- Round 1 cutoffs relative to EV_2 . Round 2 cutoffs relative to success probabilities
- Cutoffs stable across years. $\hat{\sigma}$ small $\Leftrightarrow$ distorted restricted success probabilities

Restricted Estimates with Matriculation

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.0071 (0.0026, 0.0194)	36.35% (29.47%, 43.84%)	
⋮	⋮	⋮	
2019	0.0095 (0.0035, 0.0255)	37.41% (31.04%, 44.26%)	
Sigma			0.0020** (0.0010)

- With matriculation, EV_2 's functional form yields lower Round 1 cutoffs

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Deviation Gains Test

- Suppose committee adopted model's decision rule. Would average success probability of acceptance/matriculation set have increased? If so, by how much?
 - Select applicants with model, compute average yearly success probability of model's acceptance/matriculation set, and compare to that of actual decision rule's set
 - Plug $\hat{\theta}_S^U$ into model because $\hat{\theta}_S^R$ may be distorted to rationalize suboptimal behavior
 - Compute average success probabilities across $B = 500$ perturbations $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$

Theorem 3

P-value is fraction of perturbations with model's $\overline{P(\text{Success})}$ less than committee's.

- ★ Model's set must be exogenously chosen, e.g. using $\hat{\theta}_S^U$ from a different dataset

Deviation Gains Correction

- Observed deviation gain is combination of true gain and “winner’s curse”

$$\begin{aligned} g(\hat{\theta}_S^U) &= \underbrace{\bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U)}_{\text{Model}} - \underbrace{\bar{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{Committee}} \\ &= \underbrace{\bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U) - \bar{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U)}_{\text{Winner's curse}} + \underbrace{\bar{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U) - \bar{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{True gain at } \hat{\theta}_S^U} \end{aligned}$$

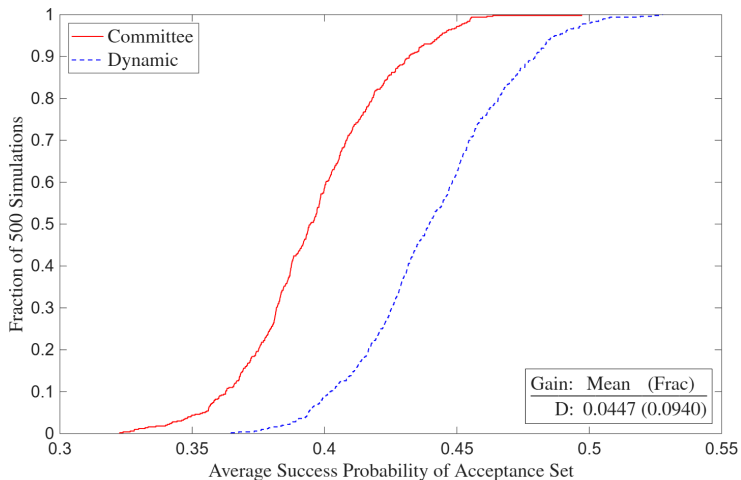
- Subtract estimate of winner’s curse from observed deviation gain

$$\widehat{\text{Winner's curse}} = \frac{1}{B} \sum_{b=1}^B [\bar{P}_{\mathcal{A}^R(\tilde{\theta}_{S,b}^U)}(\tilde{\theta}_{S,b}^U) - \bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\theta}_{S,b}^U)]$$

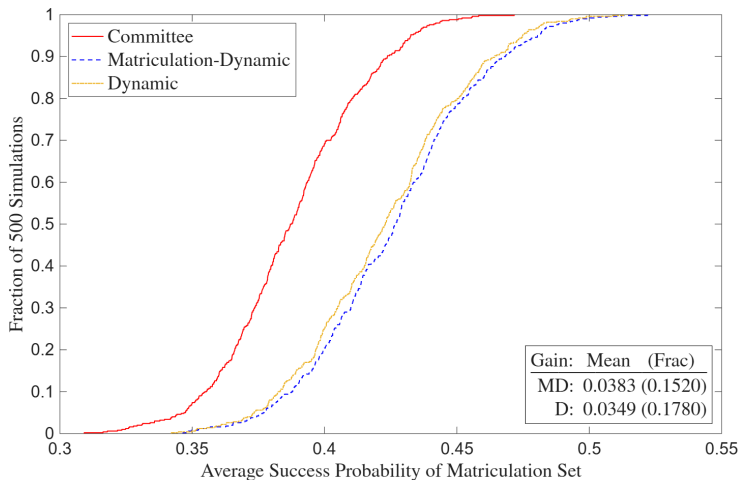
Theorem 4

Correction is exact under certain conditions and removes at least $\sqrt{2} - 1$ of the curse.

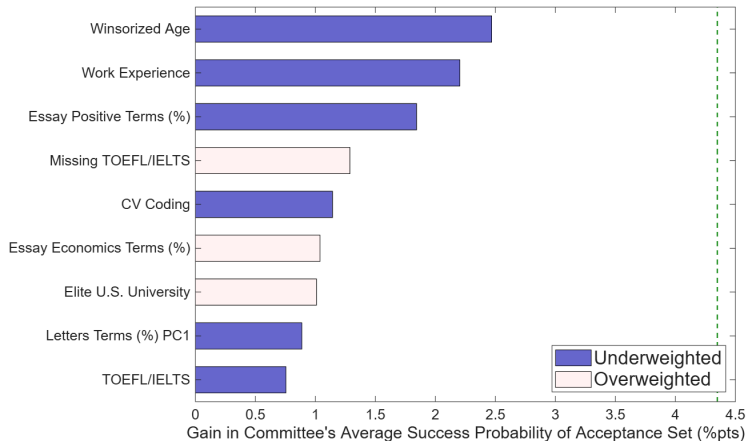
Deviation Gains (Acceptance)



Deviation Gains (Matriculation)



Deviation Gains Attribution (Acceptance)

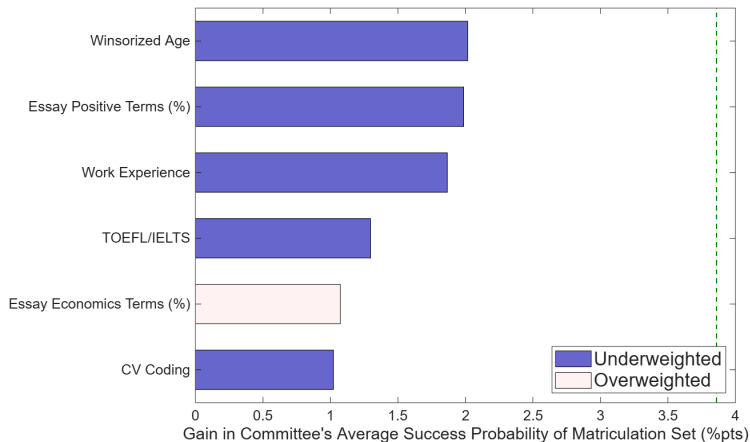


- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

» Full Deviation Gains Attribution (Acceptance)

» Deviation Gains Attribution Table (Acceptance)

Deviation Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

» Full Deviation Gains Attribution (Matriculation)

» Deviation Gains Attribution Table (Matriculation)

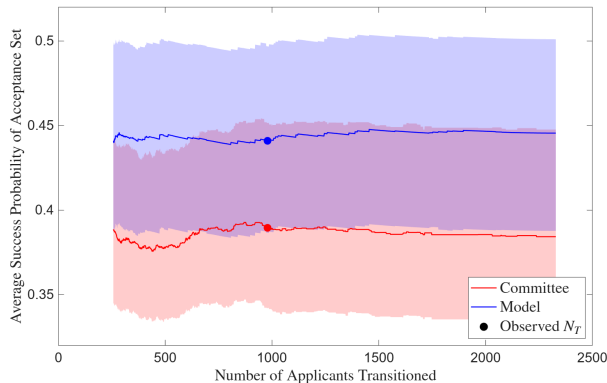
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CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

- Blue = underweighted, white = correctly weighted, red = overweighted
- AMEs don't reveal if committee deviates from optimal ranking near the cutoff

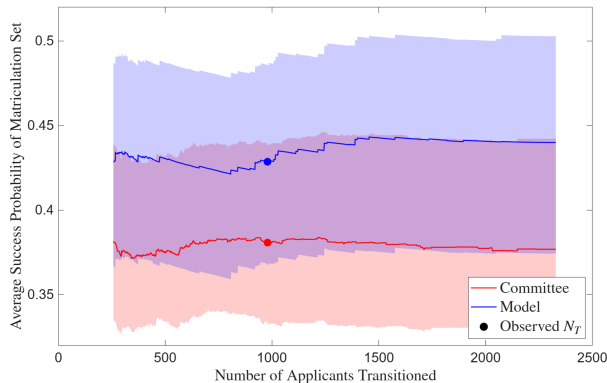
» Acceptance vs. Success AMEs

Benefit of Reading More Files (Acceptance)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_A \bar{P}(n) - cn$
- FOC: $V = c / (N_A \bar{P}'(N_T))$
- V is the dollar value of an additional expected success

Benefit of Reading More Files (Matriculation)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_M \bar{P}(n) - cn$
- FOC: $V = c / (N_M \bar{P}'(N_T))$
- V is the dollar value of an additional expected success

Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to test optimality and quantify deviation gains
 - ① For most applicants, estimated success probabilities align with those implied by committee's decisions. But some applicant characteristics are misweighted
 - ② Correctly weighting these characteristics would increase admitted cohort's average success probability by about 4 percentage points without reading more files
 - ③ Success is hard to predict, but even modest gains in selection are worthwhile
- ★ Methods apply to undergraduate/professional admissions, hiring, and other settings

Thank You!

A1: Synthetic Rows $\neq$ Actual Applicants

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	70	1	0	1	0	.	27
0	0	0	0	0	92	1	1	0	1	.	150
0	0	0	0	0	85	1	1	0	0	.	80
0	0	0	0	0	72	1	0	0	1	.	51
0	0	0	0	0	74	0	1	0	0	.	150
0	0	0	0	0	79	0	0	0	1	.	26
1	1	0	0	1	97	0	0	1	1	8	43
0	0	0	0	0	76	1	0	0	0	.	17
1	1	0	0	0	75	0	0	0	0	3	70
0	0	0	0	0	75	0	0	1	0	.	150



Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	92	0	0	1	1	8	31
0	0	0	0	0	83	0	0	0	1	.	60
0	0	0	0	1	94	0	1	0	1	.	19
1	1	1	1	0	90	0	0	1	1	5	37
0	0	0	0	1	86	0	0	1	0	.	52
1	0	1	0	1	91	1	1	0	0	2	29
1	1	0	0	0	86	0	0	0	1	5	71
1	1	0	0	0	98	0	1	0	0	5	21
1	1	0	0	1	88	1	1	0	1	4	37
0	0	0	0	0	92	1	1	0	0	.	150

A1: Dependent Variables

Admission	Success
1 (Transition)	1 (Complete Program)
<i>1 (Accept)</i>	<i>1 (Academic Placement)</i>
1 (Waitlist)	1 (Government Placement)
1 (Open House)	1 (Consulting Placement)
<i>1 (Matriculate)</i>	<i>1 (Tech Placement)</i>
1 (Attend Program)	1 (Other Placement)
Program Rank	1 (No Placement)
	Placement Rank
	1 (Top 100 Placement)

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
 - For non-tenure track, use weighted average of university's and maximum rankings
 - For non-academia, use average of ChatGPT, Claude, and Gemini's [rankings](#)

A1: Objective Variables

Demographic	Performance
Year	GRE Verbal Percentile
1 (<i>Covid Year</i>)	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
1 (Female)	Undergraduate GPA
1 (U.S.)	TOEFL/IELTS
1 (<i>U.S. African/Hispanic/Native</i>)	1 (Elite U.S. University)
1 (<i>U.S. Asian</i>)	1 (Elite U.S. Liberal Arts College)
1 (<i>U.S. Other</i>)	1 (Other U.S. University/College)
1 (International Asian)	1 (Elite Foreign University)
1 (International Other)	1 (Other Foreign University)
	1 (Graduate Degree)
	1 (Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	<i>Concentration Dummy Variables</i>

A1: Subjective Variables

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	#Math Courses
Letters Length	1 (Advanced Math)
Positive Terms (%) (YS 2012)	CV Length
Negative Terms (%) (YS 2012)	CV Coding
Standout Terms (%) (BEMS 2022)	CV 1 (Publication/Working Paper)
Ability Terms (%) (BEMS 2022)	Essay Length
Research Terms (%) (BEMS 2022)	Essay 1 (Specific Professor)
Grindstone Terms (%) (BEMS 2022)	Essay Economics Terms (%)
Teaching Terms (%) (BEMS 2022)	Essay Research Terms (%)
Communal Terms (%) (BEMS 2022)	Essay Positive Terms (%)
Agentic Terms (%) (MHM 2009)	Essay Negative Terms (%)
	Field Dummy Variables

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

A1: Dictionaries

● Recommendation Letters:

- **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop

● CVs and Essays:

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
 - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
 - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for “grindstone terms” (e.g. depend*, diligen*)
- Research questions emphasize prediction over inference
 - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
 - If GRE predicts success, committee should use it regardless of why

A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
 - Let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$
 - **Step 1:** Run probit of d on z and compute inverse Mills $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
 - **Step 2:** Run OLS or heckprobit of y on $(x, \hat{\lambda})$ to get $\hat{\beta}$
 - **Instruments:** Variables that predict matriculation but not the outcome

A1: Acceptance, Matriculation, and Success AMEs

Dependent Variable	Accept	Matriculate	T100 Placement
Winsorized Age	-0.0001	-0.0223**	0.0053*
Female	0.0163**	-0.0490	-0.0032
U.S.	-0.0013	0.1472*	0.0192
International Asian	-0.0214***	0.1644**	-0.0127
GRE Quantitative Percentile	0.0011**	-0.0147**	0.0006
Undergraduate GPA	0.0195	-0.1026	-0.1096**
TOEFL/IELTS	-0.0022	-0.0631	0.0409*
Work Experience	0.0140**	0.1591**	0.0358**
Committee Score	0.0788***	-0.0371	0.0007
Advanced Math	0.0246	-0.1950**	-0.0287
Letters Terms (%) PC1	0.0055	0.0671**	0.0159
CV Coding	-0.0044*	-0.0414**	0.0125
Essay Economics Terms (%)	0.0001	0.0366**	-0.0104
Essay Positive Terms (%)	-0.0013	-0.0072	0.0123*
Open House		0.2615***	
Program Rank			-0.0011***
Missing Program Rank			-0.3965***

Year Dummies
 GRE Verbal Percentile
 GRE Analytic Percentile
 Elite U.S. University
 Elite U.S. Liberal Arts College
 Elite Foreign University
 Other Foreign University
 Graduate Degree
 #Professor Recommenders
 #Prolific Recommenders
 #Math Courses
 Letters Length/100
 Letters Terms (%) PC2
 CV Length/100
 CV Publication/Working Paper
 Essay Length/100
 Essay Specific Professor
 Essay Research Terms (%)
 Essay Negative Terms (%)
 Missing Dummies

A1: Insufficiency of Comparing AMEs

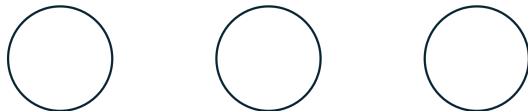
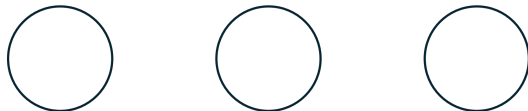
- Cannot test optimality by comparing AMEs across regressions. Suppose:
 - Work experience has large effect on success, graduate degree smaller effect
 - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
 - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
 - **Acceptance Regression:** $AME_{\text{Experience}} = 1, AME_{\text{Degree}} = 0$
 - **Success Regression:** $1 > AME_{\text{Experience}} > AME_{\text{Degree}} > 0$
 - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

A1: Diversity Objective

- **Demographics:** Squares vs. circles

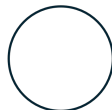
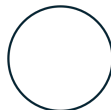
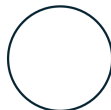
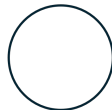


- **Quality:** High vs. low



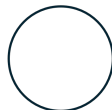
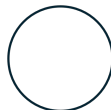
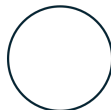
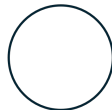
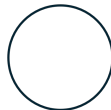
A1: Diversity Objective

- No preference for squares or circles



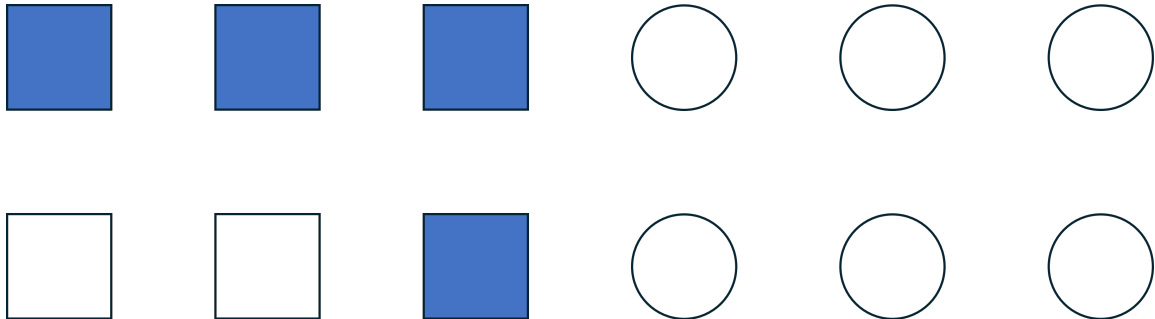
A1: Diversity Objective

- Small preference for squares $\Rightarrow$ consistent with maximizing quality



A1: Diversity Objective

- Large preference for squares $\Rightarrow$ inconsistent with maximizing quality



A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
 - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
 - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
 - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a “success” model
 - **Problem:** Not a test if admissions decisions are optimal
 - Need structural model of admissions to conclude about optimality

A1: Logistic Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$. Let $P(Y_n^E = 1|x_n^E; \theta^E) = \frac{\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}{1 + \exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}$,

and $P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE}) = \frac{\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}{1 + \exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U(y|x; \theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E; \theta^E)^{Y_n^E} [1 - P(Y_n^E = 1|x_n^E; \theta^E)]^{1-Y_n^E} \prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})^{Y_n^{NE}} [1 - P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E} \sum_{n=1}^{N_E} \hat{P}_n^E (1 - \hat{P}_n^E) \hat{\theta}_1^E - \frac{1}{N_{NE}} \sum_{n=1}^{N_{NE}} \hat{P}_n^{NE} (1 - \hat{P}_n^{NE}) \hat{\theta}_1^{NE}$. Then:
 $W = g(\hat{\theta})' [\nabla_{\theta} g(\hat{\theta}) \hat{V}(\hat{\theta}) \nabla_{\theta} g(\hat{\theta})']^{-1} g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

A1: Linear Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$, i.e. (Economics $\perp\!\!\!\perp$ Non-Economics) $|$ Predictors.

Let $f(y_n^E|x_n^E; \beta^E, \sigma_E^2) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E)^2\right]$, and

$f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE})^2\right]$. Then:

$$\mathcal{L}_N^U(y|x; \beta, \sigma^2) = \prod_{n=1}^{N_E} f(y_n^E|x_n^E; \beta^E, \sigma_E^2) \prod_{n=1}^{N_{NE}} f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2).$$

- $Y \in \{\text{Standardized Committee Score, Standardized Placement Rank}\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.)
Then $W = g(\hat{\beta}_1)'[\nabla_{\beta, \sigma^2} g(\hat{\beta}_1) \hat{V}(\hat{\beta}, \hat{\sigma}^2) \nabla_{\beta, \sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n, \sigma_n^2 = \sigma^2$)

A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
 - Need different skills at different stages in the program, such as grades vs. placement
 - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
 - **Admission:** May have different preparation and/or interests
 - **Completion:** Depending on country, may have fewer outside options
 - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
 - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
 - For economics, increased importance of elite predocs and Fed RAs

A2: Models of Admissions

- ① **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
 - ① **Static Model:** In Round 1, rank applicants based only on objective variables
 - ② **Myopic Model:** Like dynamic, but distribution of z not conditional on x
- ② **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)

A2: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z, t, \bar{k})$, where Success = $\mathbb{1}(\text{Complete Program})$ or $\mathbb{1}(\text{Good Placement})$. If reject applicant, get $P(y_2^* = S|t) = P(\text{Success})$ of year t 's marginal accepted applicant
- **Round 1:** If reject applicant, get $L/P(y_1^* = S|t) = L/P(\text{Success})$ of year t 's marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort
 - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] =$ expected value of Round 2, where z is integrated out conditional on (x, t) based on $F(z|x, t)$
 - **Static:** If transition applicant, get $P(y = S|x, t, \bar{k})$, or $P(\text{Success})$ unconditional on z . It may differ from $P(y = S|x, z, t, \bar{k})$, though it still accounts for z 's effect on y
 - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$, replace $F(z|x, t)$ with $F(z|t)$

A2: Portfolio Choice to Cutoff Rule (BEMS 2022)

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda^* \end{cases}$$

Upshot: λ^* is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2.1: Portfolio Choice to Cutoff Rule with Shocks

Portfolio Choice Problem:

$$\begin{aligned} \max_{(d_1, \dots, d_N) \in \{A, R\}^N} & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + \epsilon_n(R) \mathbb{1}(d_n = R) \}, \\ \text{such that: } & \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \dots, d_N; \lambda) = \\ & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + [\lambda + \epsilon_n(R)] \mathbb{1}(d_n = R) \} + \lambda(N_A - N) \end{aligned}$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n, \epsilon_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

Theorem 5

For large N , there is a cutoff rule solution with logistic CCPs.

► Portfolio Choice Problem

A2: Portfolio Choice to Cutoff Rule, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] + \lambda_2 (N_A - N_T)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(x_n, z_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda_2^* \end{cases}$$

Upshot: λ_2^* is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2: Portfolio Choice to Cutoff Rule, Round 1 (Dynamic)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n), \bar{\lambda}_2^*\} dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff level. Denote it hereafter as $L(y_1^* = S) \in [EV_{2,N_T+1}, EV_{2,N_T}]$

A2: Portfolio Choice to Cutoff Rule, Round 1 (Static)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S | x_n) > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S | x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S | x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in [P(y_{N_T+1} = S | x_{N_T+1}), P(y_{N_T} = S | x_{N_T})]$

A2: Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, t; \theta_A) = \frac{\exp(f_A(x, z, t)\theta_A)}{1 + \exp(f_A(x, z, t)\theta_A)}$$

Success Probability:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P(y_2^* = S | t; \theta_{S_2^*}) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

A2: Dynamic Parameterization 2

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \tilde{d}_2)/\sigma} \right) dF(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t)\end{aligned}$$

- **Continuous:** $f_{z|x,t}(z_j|x, t; \beta_{F_{z|x,t,j}}, \sigma_{F_{z|x,t,j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x,t,j}}^2}} \exp \left[\frac{-1}{2\sigma_{F_{z|x,t,j}}^2} (z_j - g(x, t)\beta_{F_{z|x,t,j}})^2 \right]$
- **Binary:** $P(z_j = 1|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}})}{1 + \exp(g(x, t)\theta_{F_{z|x,t,j}})}$
- **Multinomial:** $P(z_j = \ell|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}}^\ell)}{1 + \sum_{i=1}^{L_j-1} \exp(g(x, t)\theta_{F_{z|x,t,j}}^i)}, \ell = 1, \dots, L_j - 1$

A2: Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\
 & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Success mostly unobservable for applicants from fewer than six years ago

A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A) \mathbb{1}(d_{2,n}=A)}_{\text{Conditional choice probabilities}} \\
 & \underbrace{(1 - P_{d_{2,n}}(\theta_A)) \mathbb{1}(d_{2,n}=R)}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^N \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{f\left(z_n^{-c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{-c}|x, t} \\
 & \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{P_{y_n}(\theta_S) \mathbb{1}(y_n=S) (1 - P_{y_n}(\theta_S)) \mathbb{1}(y_n=F)}_{\text{Success probabilities}}
 \end{aligned}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

A2: Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

» Dynamic Model

A2: Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

» Dynamic Model

A2: Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S | x, t, \bar{k}) \mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

A2: Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S | x, t, \bar{k}) \mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- $P(y = S | x, t, \bar{k})$ estimates inconsistent, but fitted values account for z 's effect on y

A2: Static Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \\ \prod_{n=1}^N \underbrace{[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})]^{\mathbb{1}(y_n=S)} [(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success mostly unobservable for applicants from fewer than six years ago

A2: Cutoff Rule with Matriculation

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^N P(m_n = M | w_n) \mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | w_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] P(m_n = M | w_n) + \lambda(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(w_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda^* \\ R & \text{if } P(y_n = S | w_n) < \lambda^* \end{cases}$$

A2: Cutoff Rule with Matriculation (Large N)

Portfolio Choice Problem:

$$\max_{\delta \in \{\delta: \mathbb{R}^K \rightarrow [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w)P(m = M|w)\delta(w)f(w)dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w)\delta(w)f(w)dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} [P(y = S|w)\delta(w) + \lambda(1 - \delta(w))]P(m = M|w)f(w)dw + \lambda(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K, \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0, 1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

Upshot: Point-identified cutoff probability λ^* is with respect to success probabilities

A2: Cutoff Rule with Matriculation, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | w_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] P(m_n = M | w_n) + \lambda_2 (N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(w_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | w_n) < \lambda_2^* \end{cases}$$

A2: Cutoff Rule with Matriculation, Round 1

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n) - \bar{\lambda}_2^*, 0\} P(m_n = M | x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

A2: Matriculation-Dynamic Likelihood

Matriculation Probability:

$$P_m = P(m = M | x, z, t; \theta_M) = \frac{\exp(f_M(x, z, t)\theta_M)}{1 + \exp(f_M(x, z, t)\theta_M)}$$

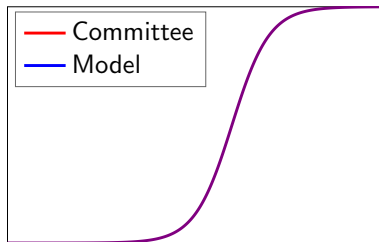
$$\begin{aligned} \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_A} \underbrace{P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)} (1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)}}_{\text{Matriculation probabilities}} \\ & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\ & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}} \end{aligned}$$

A2: Empirical Considerations

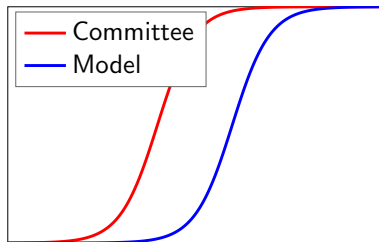
- **Selection Effects:** Track success for all applicants, including non-matriculants
 - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
 - In decision blocks containing success, fix *Program Rank* at this university's rank
 - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
 - **Example:** Recommendation letter has nuances that algorithms can't detect
 - Include subjective committee score in z as a proxy for this information

A2: How Deviation Gains Test Works

- 1 Write Bellman equations for committee's optimization problem
- 2 Estimate parameters. Rationality assumption distorts estimates
- 3 Re-estimate parameters without rationality, and use to solve model
- 4 Compare selections from model's decision rule to actual decision rule
- 5 To prevent overfitting, use perturbations of estimates for comparison



Optimal Committee



Suboptimal Committee

A2: Deviation Gains Algorithm

Dynamic/Static/Myopic:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic and myopic, or $P(y = S|x, t, \bar{k})$ for static. Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$, and accept top N_A survivors.
Compute $\overline{P(y = S|x, z, t, \bar{k})}$ of acceptance set, and compare to committee's set

Matriculation-Dynamic/Static:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \bar{k})$ for static (best analogue). Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$ and accept them in descending order, each matriculating with probability $P(m = M|x, z, t)$, until N_M matriculate.
Compute $\overline{P(y = S|x, z, t, \bar{k})}$ of matriculation set, and compare to committee's set

A2: Randomization Test

Theorem 6

Let d^R be the model's decision rule, and d^U the actual decision rule. By optimality:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1} \{ d_{1,n} [P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^R [P_{y_n}(\hat{\theta}_S^U)] = A \} \\ \geq \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1} \{ d_{1,n} [P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^U [P_{y_n}(\hat{\theta}_A^U)] = A \} \end{aligned}$$

- But for perturbation $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$'s asymptotic distribution, can have:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \tilde{\theta}_{S,b}^U) \mathbb{1} \{ d_{1,n} [P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^R [P_{y_n}(\hat{\theta}_S^U)] = A \} \\ < \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \tilde{\theta}_{S,b}^U) \mathbb{1} \{ d_{1,n} [P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^U [P_{y_n}(\hat{\theta}_A^U)] = A \} \end{aligned}$$

- For matriculation models, compare matriculants, not acceptances

A2: Deviation Gains Interpretation

- Under meaningfully-sized deviation gains, committee may have incorrect beliefs about which types of applicants are most likely to succeed
 - Too few observations for a true out-of-sample test, and cannot account for changes in the determinants of success over time, such as writing essays with AI
- Committee may use success-predicting variables that I do not observe
 - Because the success probability measure used to evaluate gains cannot detect such variables, any edge the committee's private information provides goes unmeasured
 - Comprehensive data, especially committee score in z , protects against this limitation

A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
 - **Example:** Applicants with work experience more likely to succeed and matriculate
 - Can help committee find “people that we can afford” (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
 - Instead maximize combination of $P(\text{Success})$ and other objectives
 - **Example:** Committee may want diverse cohort (e.g. across fields):
 $P(\text{Success})^\alpha \times \text{Diversity}^{1-\alpha}$, where $\alpha \in [0, 1]$ can be estimated
- **Extension 3:** Waitlist and multiyear models (see [online appendix](#))
 - Three-round models, and two-round models that repeat over infinite horizon
 - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

A3: Summary Statistics

Objective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1829	0.3873	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Winsorized Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6307	0.3026	3.6645	0.3003	3.7227	0.2610	3.7068	0.2880
TOEFL/IELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Profic Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL/IELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Winsorized Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	2329		979		257		79	

Subjective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2121
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

Regression Results

67 / 28

A3: Recommendation Letters Principal Components

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

A3: Logistic Regression Results

Dependent Variable	Asian	Asian	Hispanic	150 Percentile	150 Percentile	150 Percentile	150 Percentile	Program Rank	Program Rank	Program Rank
2016	-0.3628	-0.3628	-0.3627	0.0000	0.0000	0.0000	1.8028	1.3867	-0.8418	-0.8418
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
2017	-0.0894	-0.0894	-0.0893	0.0000	0.0000	0.0000	2.7758	1.0208	0.7671	0.7671
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
2018	-0.0894	-0.0894	-0.0893	0.0000	0.0000	0.0000	2.7758	1.0208	0.7671	0.7671
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
2019	-0.0894	-0.0894	-0.0893	0.0000	0.0000	0.0000	2.7758	1.0208	0.7671	0.7671
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Winsorized Age	0.2603	0.2603	0.2603	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Undergrad	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
U.S.	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
International Born	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
GRE Verbal Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
GRE Quantitative Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
GRE Analytic Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Undergraduate GPA	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
TOEFL/IELTS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Elite U.S. University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Elite U.S. Liberal Arts College	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Elite Foreign University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Other Foreign University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Missing Undergraduate Institution	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Graduate Degree	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Work Experience	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Committee Score	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
CV Coding	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)
Missing Program Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.2841)	(0.2841)	(0.2841)	(0.2712)	(0.2712)	(0.2712)	(0.2841)	(0.2841)	(0.2841)	(0.2841)

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A3: Logistic Regression AMEs

[illegible]

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A3: Logistic Regression Results (Synthetic)

[illegible]

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A3: Logistic Regression AMEs (Synthetic)

Dependent Variable	Age	Age ²	Gender	Married	TOEFL Pctscore	TOEFL Pctscore ²	TOEFL Pctscore ³	Program Rank	Program Rank ²	Program Rank ³
2004	-0.0001	-0.0000	0.0001	0.0000	0.0002	0.0000	0.0000	-0.0001	-0.0000	-0.0001
2007	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
2008	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
2009	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Western Age	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Female	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
U.S.	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
International Age	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
USC Virtual Presence	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
USC Quantitative Presence	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
USC Analytic Presence	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Undergraduate GPA	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
TOEFL/IELTS	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Elite U.S. University	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Elite U.S. Liberal Arts College	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Elite Foreign University	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Elite Foreign University	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Missing Undergraduate Institution	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Graduate Degree	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Work Experience	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
JPProfile Recommenders	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
JPProfile Recommenders	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Missing GRE	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Missing Undergraduate GPA	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Missing TOEFL/IELTS	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Quantitative Score	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Math Courses	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Abstract Math	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Letter Length/US	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Letter Score [%] PC2	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Letter Score [%] PC2	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
CV Length/US	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
CV Coding	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
CV Publications/Working Paper	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Essay Length/US	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Essay Specific Profound	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Essay Economic Terms [%]	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Essay Positive Terms [%]	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Essay Negative Terms [%]	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Missing Covariate Score	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Missing Application PM	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Open Hours	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Program Rank	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Missing Program Rank	-0.0001	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0000	-0.0001
Sample	50	50	Average = 1	50	50	50	50	50	50	50
Observations	2004	2007	2008	2009	2004	2007	2008	2009	2004	2007

Robust standard errors in parentheses. *p < 0.10. **p < 0.05. ***p < 0.01

- **Accept** ($p < .1$): 2017 (-), Winsorized Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success** ($p < .1$): Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A3: Lasso Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1107								
2017		-0.4928							
2018		-2.4456	0.0186						
2019		-1.7752							
Winsorized Age				0.0084	0.0047	0.0184			
Female	0.2674	0.5589							
U.S.	-0.1959						0.0477		
International Asian	-0.8686	-0.8651	0.1271				-0.5295	-0.7927	
GRE Verbal Percentile	0.0227	0.0047	-0.0015	0.0015			-0.0085		
GRE Quantitative Percentile	0.1027	0.0174	-0.0273	0.0179	0.0125		-0.2706	-0.2243	-0.0070
GRE Analytic Percentile	0.0056	0.0035							
Undergraduate GPA	0.9812	0.7135				-0.1316		0.2561	
TOEFL/IELTS	0.3055		-0.2943	0.0300		0.0934	-0.6903	-0.5850	-2.2692
Elite U.S. Liberal Arts College	0.4024	0.1112							
Elite Foreign University	-0.4025								
Other Foreign University	0.0790	0.0940	0.1998					-0.2265	
Missing Undergraduate Institution		-0.0036							
Graduate Degree	0.1178								
Work Experience	0.6006	0.4068	0.0404	0.0961	0.0470	0.1976		-2.0809	0.0922
#Prolific Recommenders	0.1534	0.0301	-0.0870	0.0526	0.0363		-0.0789	-0.0358	
Missing GRE	-0.6534			-0.0697					
Missing Undergraduate GPA	-0.2348	-0.1148							
Missing TOEFL/IELTS	0.1447			-0.1243	-0.1295	1.8402	2.2548	-0.4899	
Committee Score		3.4243			0.0353				
#Math Courses	-0.0553	0.0354							
Advanced Math	0.4302	0.7352							
Letters Length/100	0.1196	0.0587							
Letters Terms (%) PC1	0.0746	0.1485	0.1258			0.0102			
Letters Terms (%) PC2	-0.0162	-0.2901							
CV Length/100	0.0219	0.0298		0.0014		-0.0286	-0.1179		
CV Coding	-0.0038	-0.1754		0.0165	0.0076	-0.2540	-0.4154	-0.0998	
CV Publication/Working Paper	0.5563	0.6436		0.0892		-0.0687			
Essay Length/100	0.1202	0.0295							
Essay Specific Professor	0.3403	0.3958	0.0674						
Essay Economics Terms (%)	-0.0016			-0.0405	-0.0325	-0.0104	0.4473	0.6570	0.2117
Essay Research Terms (%)		-0.0278						-0.1506	
Essay Positive Terms (%)	-0.0049	0.0382		0.0192	0.0140	0.0267			
Essay Negative Terms (%)		-0.0417							
Missing Committee Score		-0.8416			-0.2615	-0.0714	4.0146	0.7071	
Missing Application Pdf	-1.0365								
Open House			1.0982						
Program Rank						-0.0082		0.1385	
Missing Program Rank						-2.2557		19.4273	
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R ²	0.2841	0.8007	0.1281	0.0196	0.0213	0.2265	0.0206	0.0278	0.2693
CV Pseudo-R ²	0.2441	0.7605	0.0596	0.0062	0.0129	0.2190	0.0121	0.0183	0.2638

- Accept:** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Verbal Percentile (+), GRE Quantitative Percentile (+), GRE Analytic Percentile (+), Undergraduate GPA (+), Elite U.S. Liberal Arts College (+), Elite Foreign University (-), Other Foreign University (+), Work Experience (+), # Prolific Recommenders (+), Missing Undergraduate GPA (-), Committee Score (+), #Math Courses (-), Advanced Math (+), Letters Length/100 (+), Letters Terms (%) PC1 (+), Letters Terms (%) PC2 (-), CV Length/100 (+), CV Coding (-), CV Publication/Working Paper (+), Essay Length/100 (+), Essay Specific Professor (+), Economics Research Terms (%) (-), Essay Positive Terms (%) (-), Essay Negative Terms (%) (-), Missing Committee Score (-)
- Matriculate:** 2018 (+), International Asian (+), GRE Verbal Percentile (-), GRE Quantitative Percentile (-), TOEFL/IELTS (-), Other Foreign University (+), Missing Undergraduate Institution (-), Work Experience (+), #Prolific Recommenders (-), Letters Terms (%) PC1 (+), Essay Specific Professor (+), Open House (+)
- T100 Placement:** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Letters Terms (%) PC1 (+), Essay Economics Terms (%) (-), Essay Positive Terms (%) (+), Missing Committee Score (-), Program Rank (-), Missing Program Rank (-)

Restricted

[illegible]

A3: Restricted Static Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	29.36% (23.20%, 36.38%)	39.20% (32.75%, 46.05%)	
⋮	⋮	⋮	
2019	34.04% (28.67%, 39.85%)	39.62% (32.54%, 47.16%)	
Sigma			0.0064 (0.0035)

- Static model has interpretable Round 1 cutoff probabilities
- Round 2 cutoff probabilities always greater than those in Round 1

Restricted

74 / 28

A3: Myopic Estimates

Unrestricted

	Correlation Score			Accept			Reject		
	Coefficient	Test Statistic	p-Value	Coefficient	Test Statistic	p-Value	Coefficient	Test Statistic	p-Value
Intercept									
(0.000)	2.088	(0.235)	0.882	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.1761	(2.088)	0.818	0.415	0.261	(0.261)	0.581	0.272	(0.272)	0.786
(0.000)	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
2017	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
2018	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
2019	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Unusual Age	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
U.S.	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Female	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
International Area	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
GRI Verbal Personality	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
GRI Quantitative Personality	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
GRI Analytic Personality	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
TOP3, US, BELTS	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
U.S. LIBERAL ARTS COLLEGE	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Other Foreign University	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Mixing Undergraduate Institution	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Graduate Degree	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Black Experience	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Hispanic Experience	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Mixing GRI	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Mixing TOP3, BELTS	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Commitment Score	0.1761	(0.000)	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
0.4333*	(0.000)	0.1761	0.504	0.261	(0.261)	0.581	0.272	(0.272)	0.786
Math Courses	-0.0004	(0.9984)	0.9996	-0.0004	(0.9984)	0.9996	-0.0004	(0.9984)	0.9996
Advanced Math	-0.0003	(0.9993)	0.9997	-0.0003	(0.9993)	0.9997	-0.0003	(0.9993)	0.9997
0.9987	(0.0000)	0.9993	0.9997	-0.0003	(0.9993)	0.9997	-0.0003	(0.9993)	0.9997
Letters Length 100	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999
Letters Length 100 % F12	-0.0024	(0.9991)	0.9999	-0.0024	(0.9991)	0.9999	-0.0024	(0.9991)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0024	(0.9991)	0.9999	-0.0024	(0.9991)	0.9999
Letters Length 100 % F12	-0.0036	(0.9986)	0.9999	-0.0036	(0.9986)	0.9999	-0.0036	(0.9986)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0036	(0.9986)	0.9999	-0.0036	(0.9986)	0.9999
Cy Length 100	-0.0007	(0.9999)	0.9999	-0.0007	(0.9999)	0.9999	-0.0007	(0.9999)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0007	(0.9999)	0.9999	-0.0007	(0.9999)	0.9999
Cy Coding	-0.0022	(0.9993)	0.9999	-0.0022	(0.9993)	0.9999	-0.0022	(0.9993)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0022	(0.9993)	0.9999	-0.0022	(0.9993)	0.9999
Cy Publication/Writing Paper	-0.0027	(0.9991)	0.9999	-0.0027	(0.9991)	0.9999	-0.0027	(0.9991)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0027	(0.9991)	0.9999	-0.0027	(0.9991)	0.9999
English Length 100	-0.0019	(0.9997)	0.9999	-0.0019	(0.9997)	0.9999	-0.0019	(0.9997)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0019	(0.9997)	0.9999	-0.0019	(0.9997)	0.9999
English Specific Paper	-0.0027	(0.9991)	0.9999	-0.0027	(0.9991)	0.9999	-0.0027	(0.9991)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0027	(0.9991)	0.9999	-0.0027	(0.9991)	0.9999
English Economic Times (%)	-0.0024	(0.9991)	0.9999	-0.0024	(0.9991)	0.9999	-0.0024	(0.9991)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0024	(0.9991)	0.9999	-0.0024	(0.9991)	0.9999
English Research Times (%)	-0.0004	(0.9998)	0.9999	-0.0004	(0.9998)	0.9999	-0.0004	(0.9998)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0004	(0.9998)	0.9999	-0.0004	(0.9998)	0.9999
English Practice Times (%)	-0.0019	(0.9997)	0.9999	-0.0019	(0.9997)	0.9999	-0.0019	(0.9997)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0019	(0.9997)	0.9999	-0.0019	(0.9997)	0.9999
English Negative Times (%)	-0.0028	(0.9989)	0.9999	-0.0028	(0.9989)	0.9999	-0.0028	(0.9989)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0028	(0.9989)	0.9999	-0.0028	(0.9989)	0.9999
Mixing Commitment Score	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999
Using Admission Pdf	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999
Program Rank	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999
Mixing Program Rank	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999
0.9999	(0.0000)	0.9999	0.9999	-0.0009	(0.9999)	0.9999	-0.0009	(0.9999)	0.9999

	(0.0001)
Obs./Param. LL AIC	2120 207 -10470.7 20135.4

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Restricted

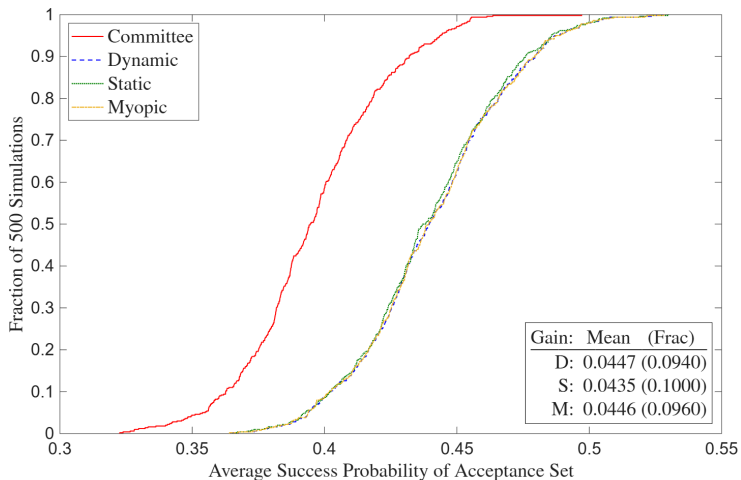
	Continuation Score	Case#1	Case#2	Case#3	Case#4	Case#5	Case#6	Case#7	Case#8	Case#9	Case#10	Case#11	Case#12	Case#13	Case#14	Case#15	Case#16	Case#17	Case#18	Case#19	Case#20	Case#21	Case#22	Case#23	Case#24	Case#25	Case#26	Case#27	Case#28	Case#29	Case#30	Case#31	Case#32	Case#33	Case#34	Case#35	Case#36	Case#37	Case#38	Case#39	Case#40	Case#41	Case#42	Case#43	Case#44	Case#45	Case#46	Case#47	Case#48	Case#49	Case#50	Case#51	Case#52	Case#53	Case#54	Case#55	Case#56	Case#57	Case#58	Case#59	Case#60	Case#61	Case#62	Case#63	Case#64	Case#65	Case#66	Case#67	Case#68	Case#69	Case#70	Case#71	Case#72	Case#73	Case#74	Case#75	Case#76	Case#77	Case#78	Case#79	Case#80	Case#81	Case#82	Case#83	Case#84	Case#85	Case#86	Case#87	Case#88	Case#89	Case#90	Case#91	Case#92	Case#93	Case#94	Case#95	Case#96	Case#97	Case#98	Case#99	Case#100	Case#101	Case#102	Case#103	Case#104	Case#105	Case#106	Case#107	Case#108	Case#109	Case#110	Case#111	Case#112	Case#113	Case#114	Case#115	Case#116	Case#117	Case#118	Case#119	Case#120	Case#121	Case#122	Case#123	Case#124	Case#125	Case#126	Case#127	Case#128	Case#129	Case#130	Case#131	Case#132	Case#133	Case#134	Case#135	Case#136	Case#137	Case#138	Case#139	Case#140	Case#141	Case#142	Case#143	Case#144	Case#145	Case#146	Case#147	Case#148	Case#149	Case#150	Case#151	Case#152	Case#153	Case#154	Case#155	Case#156	Case#157	Case#158	Case#159	Case#160	Case#161	Case#162	Case#163	Case#164	Case#165	Case#166	Case#167	Case#168	Case#169	Case#170	Case#171	Case#172	Case#173	Case#174	Case#175	Case#176	Case#177	Case#178	Case#179	Case#180	Case#181	Case#182	Case#183	Case#184	Case#185	Case#186	Case#187	Case#188	Case#189	Case#190	Case#191	Case#192	Case#193	Case#194	Case#195	Case#196	Case#197	Case#198	Case#199	Case#200	Case#201	Case#202	Case#203	Case#204	Case#205	Case#206	Case#207	Case#208	Case#209	Case#210	Case#211	Case#212	Case#213	Case#214	Case#215	Case#216	Case#217	Case#218	Case#219	Case#220	Case#221	Case#222	Case#223	Case#224	Case#225	Case#226	Case#227	Case#228	Case#229	Case#230	Case#231	Case#232	Case#233	Case#234	Case#235	Case#236	Case#237	Case#238	Case#239	Case#240	Case#241	Case#242	Case#243	Case#244	Case#245	Case#246	Case#247	Case#248	Case#249	Case#250	Case#251	Case#252	Case#253	Case#254	Case#255	Case#256	Case#257	Case#258	Case#259	Case#260	Case#261	Case#262	Case#263	Case#264	Case#265	Case#266	Case#267	Case#268	Case#269	Case#270	Case#271	Case#272	Case#273	Case#274	Case#275	Case#276	Case#277	Case#278	Case#279	Case#280	Case#281	Case#282	Case#283	Case#284	Case#285	Case#286	Case#287	Case#288	Case#289	Case#290	Case#291	Case#292	Case#293	Case#294	Case#295	Case#296	Case#297	Case#298	Case#299	Case#300	Case#301	Case#302	Case#303	Case#304	Case#305	Case#306	Case#307	Case#308	Case#309	Case#310	Case#311	Case#312	Case#313	Case#314	Case#315	Case#316	Case#317	Case#318	Case#319	Case#320	Case#321	Case#322	Case#323	Case#324	Case#325	Case#326	Case#327	Case#328	Case#329	Case#330	Case#331	Case#332	Case#333	Case#334	Case#335	Case#336	Case#337	Case#338	Case#339	Case#340	Case#341	Case#342	Case#343	Case#344	Case#345	Case#346	Case#347	Case#348	Case#349	Case#350	Case#351	Case#352	Case#353	Case#354	Case#355	Case#356	Case#357	Case#358	Case#359	Case#360	Case#361	Case#362	Case#363	Case#364	Case#365	Case#366	Case#367	Case#368	Case#369	Case#370	Case#371	Case#372	Case#373	Case#374	Case#375	Case#376	Case#377	Case#378	Case#379	Case#380	Case#381	Case#382	Case#383	Case#384	Case#385	Case#386	Case#387	Case#388	Case#389	Case#390	Case#391	Case#392	Case#393	Case#394	Case#395	Case#396	Case#397	Case#398	Case#399	Case#400	Case#401	Case#402	Case#403	Case#404	Case#405	Case#406	Case#407	Case#408	Case#409	Case#410	Case#411	Case#412	Case#413	Case#414	Case#415	Case#416	Case#417	Case#418	Case#419	Case#420	Case#421	Case#422	Case#423	Case#424	Case#425	Case#426	Case#427	Case#428	Case#429	Case#430	Case#431	Case#432	Case#433	Case#434	Case#435	Case#436	Case#437	Case#438	Case#439	Case#440	Case#441	Case#442	Case#443	Case#444	Case#445	Case#446	Case#447	Case#448	Case#449	Case#450	Case#451	Case#452	Case#453	Case#454	Case#455	Case#456	Case#457	Case#458	Case#459	Case#460	Case#461	Case#462	Case#463	Case#464	Case#465	Case#466	Case#467	Case#468	Case#469	Case#470	Case#471	Case#472	Case#473	Case#474	Case#475	Case#476	Case#477	Case#478	Case#479	Case#480	Case#481	Case#482	Case#483	Case#484	Case#485	Case#486	Case#487	Case#488	Case#489	Case#490	Case#491	Case#492	Case#493	Case#494	Case#495	Case#496	Case#497	Case#498	Case#499	Case#500	Case#501	Case#502	Case#503	Case#504	Case#505	Case#506	Case#507	Case#508	Case#509	Case#510	Case#511	Case#512	Case#513	Case#514	Case#515	Case#516	Case#517	Case#518	Case#519	Case#520	Case#521	Case#522	Case#523	Case#524	Case#525	Case#526	Case#527	Case#528	Case#529	Case#530	Case#531	Case#532	Case#533	Case#534	Case#535	Case#536	Case#537	Case#538	Case#539	Case#540	Case#541	Case#542	Case#543	Case#544	Case#545	Case#546	Case#547	Case#548	Case#549	Case#550	Case#551	Case#552	Case#553	Case#554	Case#555	Case#556	Case#557	Case#558	Case#559	Case#560	Case#561	Case#562	Case#563	Case#564	Case#565	Case#566	Case#567	Case#568	Case#569	Case#570	Case#571	Case#572	Case#573	Case#574	Case#575	Case#576	Case#577	Case#578	Case#579	Case#580	Case#581	Case#582	Case#583	Case#584	Case#585	Case#586	Case#587	Case#588	Case#589	Case#590	Case#591	Case#592	Case#593	Case#594	Case#595	Case#596	Case#597	Case#598	Case#599	Case#600	Case#601	Case#602	Case#603	Case#604	Case#605	Case#606	Case#607	Case#608	Case#609	Case#610	Case#611	Case#612	Case#613	Case#614	Case#615	Case#616	Case#617	Case#618	Case#619	Case#620	Case#621	Case#622	Case#623	Case#624	Case#625	Case#626	Case#627	Case#628	Case#629	Case#630	Case#631	Case#632	Case#633	Case#634	Case#635	Case#636	Case#637	Case#638	Case#639	Case#640	Case#641	Case#642	Case#643	Case#644	Case#645	Case#646	Case#647	Case#648	Case#649	Case#650	Case#651	Case#652	Case#653	Case#654	Case#655	Case#656	Case#657	Case#658	Case#659	Case#660	Case#661	Case#662	Case#663	Case#664	Case#665	Case#666	Case#667	Case#668	Case#669	Case#670	Case#671	Case#672	Case#673	Case#674	Case#675	Case#676	Case#677	Case#678	Case#679	Case#680	Case#681	Case#682	Case#683	Case#684	Case#685	Case#686	Case#687	Case#688	Case#689	Case#690	Case#691	Case#692	Case#693	Case#694	Case#695	Case#696	Case#697	Case#698	Case#699	Case#700	Case#701	Case#702	Case#703	Case#704	Case#705	Case#706	Case#707	Case#708	Case#709	Case#710	Case#711	Case#712	Case#713	Case#714	Case#715	Case#716	Case#717	Case#718	Case#719	Case#720	Case#721	Case#722	Case#723	Case#724	Case#725	Case#726	Case#727	Case#728	Case#729	Case#730	Case#731	Case#732	Case#733	Case#734	Case#735	Case#736	Case#737	Case#738	Case#739	Case#740	Case#741	Case#742	Case#743	Case#744	Case#745	Case#746	Case#747	Case#748	Case#749	Case#750	Case#751	Case#752	Case#753	Case#754	Case#755	Case#756	Case#757	Case#758	Case#759	Case#760	Case#761	Case#762	Case#763	Case#764	Case#765	Case#766	Case#767	Case#768	Case#769	Case#770	Case#771	Case#772	Case#773	Case#774	Case#775	Case#776	Case#777	Case#778	Case#779	Case#780	Case#781	Case#782	Case#783	Case#784	Case#785	Case#786	Case#787	Case#788	Case#789	Case#790	Case#791	Case#792	Case#793	Case#794	Case#795	Case#796	Case#797	Case#798	Case#799	Case#800	Case#801	Case#802	Case#803	Case#804	Case#805	Case#806	Case#807	Case#808	Case#809	Case#810	Case#811	Case#812	Case#813	Case#814	Case#815	Case#816	Case#817	Case#818	Case#819	Case#820	Case#821	Case#822	Case#823	Case#824	Case#825	Case#826	Case#827	Case#828	Case#829	Case#830	Case#831	Case#832	Case#833	Case#834	Case#835	Case#836	Case#837	Case#838	Case#839	Case#840	Case#841	Case#842	Case#843	Case#844	Case#845	Case#846	Case#847	Case#848	Case#849	Case#850	Case#851	Case#852	Case#853	Case#854	Case#855	Case#856	Case#857	Case#858	Case#859	Case#860	Case#861	Case#862	Case#863	Case#864	Case#865	Case#866	Case#867	Case#868	Case#869	Case#870	Case#871	Case#872	Case#873	Case#874	Case#875	Case#876	Case#877	Case#878	Case#879	Case#880	Case#881	Case#882	Case#883	Case#884	Case#885	Case#886	Case#887	Case#888	Case#889	Case#890	Case#891	Case#892	Case#893	Case#894	Case#895	Case#896	Case#897	Case#898	Case#899	Case#900	Case#901	Case#902	Case#903	Case#904	Case#905	Case#906	Case#907	Case#908	Case#909	Case#910	Case#911	Case#912	Case#913	Case#914	Case#915	Case#916	Case#917	Case#918	Case#919	Case#920	Case#921	Case#922	Case#923	Case#924	Case#925	Case#926	Case#927	Case#928	Case#929	Case#930	Case#931	Case#932	Case#933	Case#934	Case#935	Case#936	Case#937	Case#938	Case#939	Case#940	Case#941	Case#942	Case#943	Case#944	Case#945	Case#946	Case#947	Case#948	Case#949	Case#950	Case#951	Case#952	Case#953	Case#954	Case#955	Case#956	Case#957	Case#958	Case#959	Case#960	Case#961	Case#962	Case#963	Case#964	Case#965	Case#966	Case#967	Case#968	Case#969	Case#970	Case#971	Case#972	Case#973	Case#974	Case#975	Case#976	Case#977	Case#978	Case#979	Case#980	Case#981	Case#982	Case#983	Case#984	Case#985	Case#986	Case#987	Case#988	Case#989	Case#990	Case#991	Case#992	Case#993	Case#994	Case#995	Case#996	Case#997	Case#998	Case#999	Case#1000
average	0.0707***	0.0717***	0.076	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.0627***	0.062																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											

Values in parentheses are standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

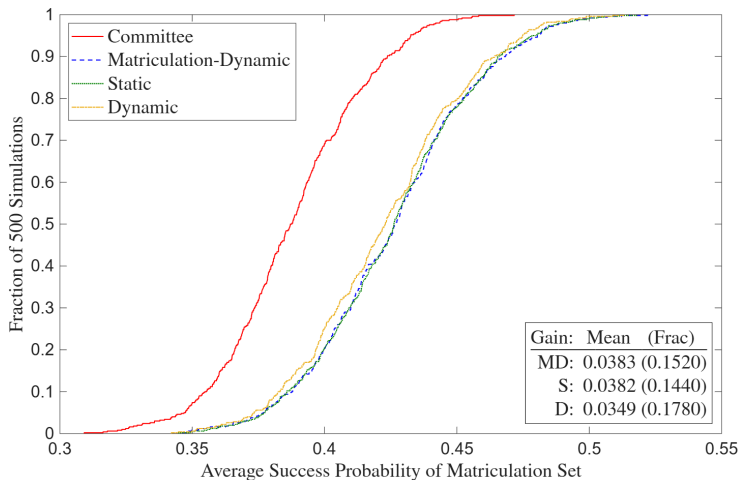
Restricted

	Companion Score	Case 1	1[C(0.6)]	Case 2	2[P(0.6)]	Score	Maximum
Keywords	0.92(0.97)	0.81(1.01)	0.00(1)	0.86(0.97)	36.3(5)	0.02(0.7)	0.85(6)
		0.30(0.5)	0.00(0.0)	0.30(0.5)	29.4(5)	0.00(0.0)	0.30(5)
	0.51(2)	0.01(1)	0.00(1)	0.00(1)	38.7(5)	0.01(1)	1.20(7)
	0.19(0)	0.18(0)	0.00(1)	0.00(0)	37.9(4)	0.00(0)	0.19(0)
2017	0.84(0.94)	0.89(0.9)	0.02(0)	0.93(0.94)	38.0(5)	0.04(0)	0.90(8)
	0.17(0)	0.17(0)	0.00(1)	0.10(0)	39.4(5)	0.00(0)	0.05(1)
2018	0.80(0.94)	0.82(0)	0.00(0)	0.80(0.94)	38.1(5)	0.00(0)	0.80(8)
	0.19(0)	0.19(0)	0.00(0)	0.00(0)	37.9(4)	0.00(0)	0.19(0)
2019	0.88(0.94)	0.82(0)	0.00(0)	0.88(0.94)	37.4(5)	0.00(0)	0.88(7)
	0.17(0)	0.17(0)	0.00(0)	0.00(0)	38.4(5)	0.00(0)	0.17(0)
Worstcase Avg	0.85(0.94)	0.85(0.94)	0.00(0)	0.85(0.94)	38.2(5)	0.00(0)	0.85(2)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Final	0.87(0)	0.87(0)	0.00(0)	0.87(0)	37.9(4)	0.00(0)	0.87(2)
	0.17(0)	0.17(0)	0.00(0)	0.17(0)	38.7(5)	0.00(0)	0.17(0)
U.S.	0.17(0)	0.17(0)	0.00(0)	0.17(0)	38.7(5)	0.00(0)	0.17(0)
	0.17(0)	0.17(0)	0.00(0)	0.17(0)	38.7(5)	0.00(0)	0.17(0)
International Rate	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
CSE Visual Processes	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
CSE Cognitive Processes	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
CSE Analysis Processes	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
TCSP, WLTS	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Elite U.S. University	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
U.S. U.S. Liberal Arts College	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Elite Foreign University	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Other Foreign University	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Abstract Syntax	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Logic Expression	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
9-Preface Recommendations	0.88(0.94)	0.88(0.94)	0.00(0)	0.88(0.94)	37.9(4)	0.00(0)	0.88(7)
	0.19(0)	0.19(0)	0.00(0)	0.19(0)	37.9(4)	0.00(0)	0.19(0)
Meaning Understanding Indicator	0.88(0.94)	0.88(0.94)	0.00(0)				

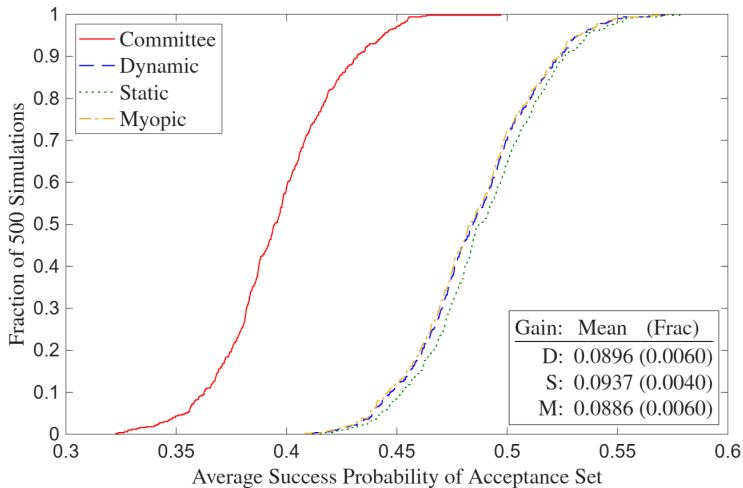
A3: Full Deviation Gains (Acceptance)



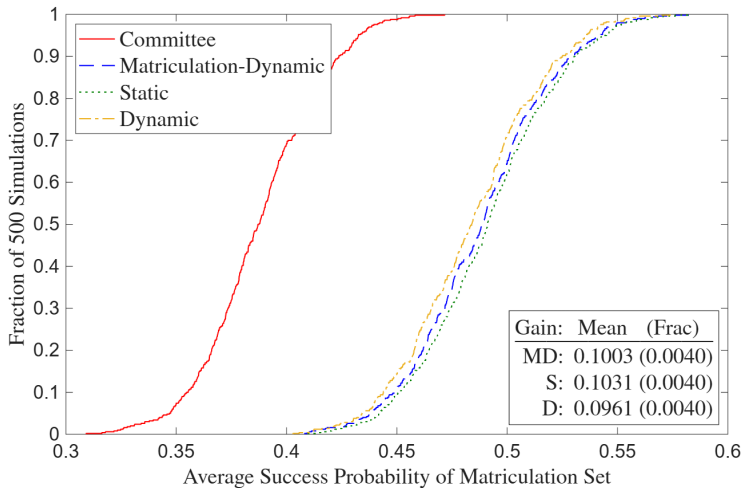
A3: Full Deviation Gains (Matriculation)



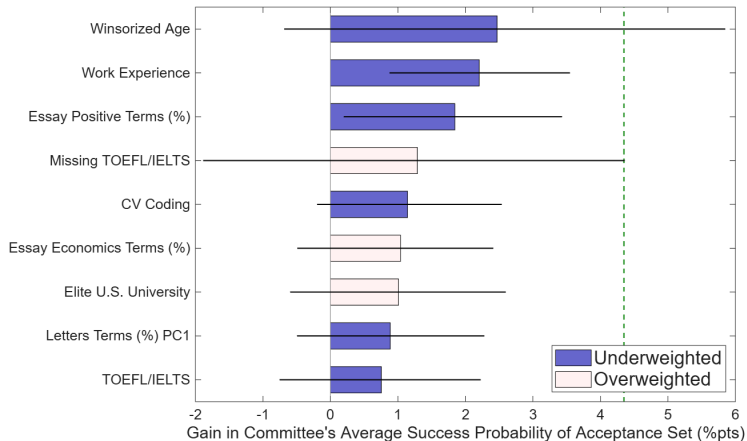
A3: Uncorrected Deviation Gains (Acceptance)



A3: Uncorrected Deviation Gains (Matriculation)

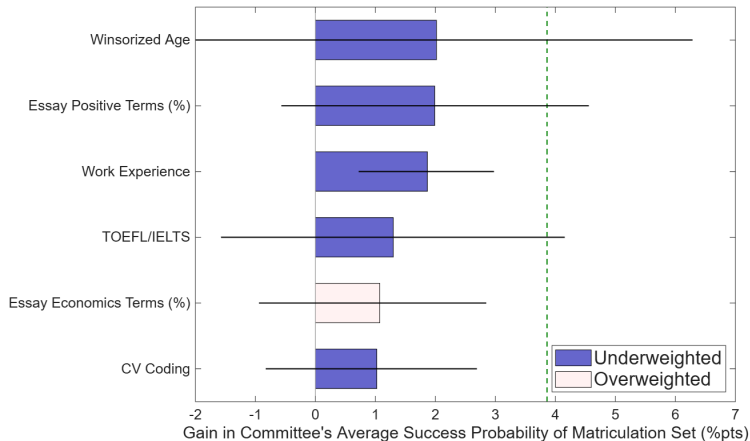


A3: Full Deviation Gains Attribution (Acceptance)



- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

A3: Full Deviation Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

A3: Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.4692	25.9910	26.2840	29.9572	29.5331	1.1674	1.0326
Work Experience	2.2037	0.5247	0.7276	0.9844	0.9650	0.5141	0.4752
Essay Positive Terms (%)	1.8431	11.7499	11.6521	12.5376	12.1183	0.7858	0.4137
Missing TOEFL/IELTS	1.2872	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.1426	2.8001	2.7820	3.3399	2.9415	0.6244	0.1785
Essay Economics Terms (%)	1.0382	9.7228	9.7423	9.0284	9.5388	-0.6411	-0.1828
Elite U.S. University	1.0086	0.1267	0.2335	0.0078	0.0389	-0.6784	-0.5848
Letters Terms (%) PC1	0.8868	-0.0106	0.1374	0.4969	0.1387	0.5202	0.0018
TOEFL/IELTS	0.7541	7.4376	7.5675	7.7759	7.6827	0.5957	0.3294

» Deviation Gains Attribution

A3: Matriculation-Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.0176	25.9910	26.3284	30.9159	29.7737	1.4580	1.0950
Essay Positive Terms (%)	1.9874	11.7499	11.6107	12.9136	12.2763	1.1534	0.5905
Work Experience	1.8659	0.5247	0.7938	0.9949	0.9799	0.4077	0.3725
TOEFL/IELTS	1.2973	7.4376	7.4939	7.8501	7.6128	1.0238	0.3397
Essay Economics Terms (%)	1.0726	9.7228	9.9994	9.0856	9.5738	-0.8271	-0.3821
CV Coding	1.0219	2.8001	2.6661	3.3395	2.8782	0.7618	0.2374

» Deviation Gains Attribution (Matriculation)

A3: Deviation Gains Robustness

	Acceptance Set	Matriculation Set
Observed Gain, Uncorrected	0.1015	0.1124
Observed Gain, Corrected	0.0514	0.0475
Perturbed Gain, Uncorrected	0.0937	0.1031
Perturbed Gain, Corrected	0.0435	0.0382
Likelihood-Ratio P-Value	0.1166	0.1166
Calibrated Signal Scale	0.4197	0.4197
Calibrated In-Sample Gain	0.1242	0.1481
Calibrated True Gain	0.0491	0.0560

►► Deviation Gains (Acceptance)

►► Deviation Gains (Matriculation)

- 1 Scale down (x, z) parameters with values of scaling factor τ from 0 to 1
- 2 Simulate *Success*, and find specific τ yielding observed joint LR test statistic
- 3 Calculate true deviation gain for that τ

A4: Graduate School Means by Program

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Means by Program, Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREa	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

► Insufficiency of Comparing AMEs

A4: Graduate School Logistic Regression Results

Dependent Variable ← Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966** (0.2010)	-0.4332* (0.2612)	-0.5752* (0.3479)	-0.5501 (0.4129)	-0.1766 (0.5545)
Winsorized Age	-0.0095*** (0.0358)	0.0454* (0.0271)	0.0380 (0.0327)	0.0222 (0.0532)	-0.0520 (0.0890)
Female	0.4079** (0.1681)	0.6723*** (0.2299)	0.6539** (0.2671)	0.1311 (0.3452)	-0.5060 (0.5606)
U.S. African/Hispanic/Native	0.4221 (0.5258)	1.0081*** (0.3545)	1.1601*** (0.4426)	0.0120 (0.8411)	-0.8263 (1.1611)
U.S. Asian	-0.5304 (0.4199)	-0.5669 (0.6504)	-0.3260 (1.2512)	0.8638 (0.7201)	1.1711 (0.7276)
International Asian	-0.8445*** (0.2945)	0.1353 (0.4146)	0.9405 (0.7233)	0.4990 (0.6729)	-0.7827 (0.9323)
International Other	0.5034 (0.3172)	0.3781 (0.4131)	1.3523*** (0.5086)	-0.1061 (0.7483)	0.6239 (1.1070)
GRE Verbal Percentile	0.0155*** (0.0060)	0.0281* (0.0160)	0.0753** (0.0344)	-0.0092 (0.0154)	0.0607*** (0.0218)
GRE Quantitative Percentile	0.1547*** (0.0186)	0.0161* (0.0086)	0.0210* (0.0116)	0.0211 (0.0150)	-0.0138 (0.0159)
GRE Analytic Percentile	0.0179*** (0.0042)	0.0199** (0.0094)	0.0030 (0.0134)	0.0175 (0.0127)	-0.0202 (0.0218)
Undergraduate GPA	2.2212*** (0.7219)	0.6968 (0.5476)	1.3648** (0.5513)	1.2545 (0.9703)	1.7772* (0.9487)
TOEFL/IELTS	0.3878 (0.2883)	0.6456 (0.4708)	1.0963** (0.5526)	0.9655* (0.5456)	-0.4656 (0.6112)
Elite U.S. University	0.8850*** (0.2960)	0.3943 (0.3136)	0.6541* (0.3022)	-0.6057 (0.4065)	1.4303** (0.6102)
Elite U.S. Liberal Arts	1.1173*** (0.3813)	0.6276* (0.3754)	1.0690** (0.5390)	-0.6088 (1.4186)	2.3151** (0.6422)
Elite Foreign University	1.0061* (0.6060)	0.4750 (0.8224)	1.7715** (0.7938)	1.6195* (0.8587)	0.3889 (1.9889)
Other Foreign University	0.5442 (0.5765)	0.5777 (0.7686)	1.3732** (0.6902)	0.4188 (0.7157)	-1.0737 (1.3016)
Graduate Degree	0.0467 (0.2238)	0.0733 (0.2583)	0.3128 (0.3280)	0.4049 (0.4806)	0.4892 (0.6399)
Work Experience	0.3932** (0.1880)	0.4027 (0.2993)	0.4972 (0.3295)	0.1495 (0.3778)	1.8201** (0.8564)
#Professor Recommenders	0.0340 (0.1317)	-0.0179 (0.1339)	0.5511** (0.2554)	0.4781* (0.2662)	-0.0981 (0.2660)
#Prolific Recommenders	0.2635*** (0.0854)	0.3025** (0.1262)	0.1880 (0.1576)	0.3284** (0.1544)	0.5901* (0.3145)
#Math Courses	0.1809*** (0.0596)				
Advanced Math	0.2319 (0.1790)				
Missing GRE	0.1899 (0.7646)	-0.3259 (0.3109)	0.6282 (0.4275)	-0.0831 (0.3761)	-1.1514** (0.5702)
Missing Undergraduate GPA	-0.4745 (0.4801)	-0.4917 (0.5440)	-1.5402** (0.6893)	-1.1097* (0.6481)	0.4660 (1.1831)
Missing TOEFL/IELTS	-0.3786 (0.2432)	0.6188 (0.5657)	0.6745 (0.5190)	-0.0440 (0.5099)	-1.7224* (0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2677	0.1525	0.1862	0.1871	0.2475

H robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

A4: Graduate School Logistic Regression AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303** (0.0153)	-0.0164 (0.0100)	-0.0431* (0.0261)	-0.0298 (0.0224)	-0.0075 (0.0238)	-0.0218*** (0.0084)
Winsorized Age	-0.0076*** (0.0027)	0.0017* (0.0010)	0.0028 (0.0025)	0.0012 (0.0029)	-0.0022 (0.0038)	0.0016* (0.0009)
Female	0.0312** (0.0126)	0.0254*** (0.0088)	0.0490** (0.0220)	0.0071 (0.0187)	-0.0216 (0.0229)	0.0207*** (0.0075)
U.S. African/Hispanic/Native	0.0323 (0.0402)	0.0381*** (0.0137)	0.0869*** (0.0332)	0.0007 (0.0455)	-0.0352 (0.0487)	0.0384*** (0.0119)
U.S. Asian	-0.0413 (0.0322)	-0.0215 (0.0247)	-0.0244 (0.0936)	0.0468 (0.0391)	0.0499 (0.0313)	0.0062 (0.0185)
International Asian	-0.0649*** (0.0224)	-0.0070 (0.0157)	0.0704 (0.0543)	0.0265 (0.0363)	-0.0334 (0.0393)	0.0069 (0.0132)
International Other	0.0385 (0.0243)	0.0143 (0.0157)	0.1013*** (0.0385)	-0.0057 (0.0406)	0.0266 (0.0471)	0.0238* (0.0132)
GRE Verbal Percentile	0.0012*** (0.0005)	0.0011* (0.0006)	0.0056** (0.0026)	-0.0005 (0.0008)	0.0026*** (0.0010)	0.0015*** (0.0005)
GRE Quantitative Percentile	0.0118*** (0.0014)	0.0006* (0.0003)	0.0016* (0.0009)	0.0011 (0.0008)	-0.0006 (0.0007)	0.0006** (0.0003)
GRE Analytic Percentile	0.0014*** (0.0003)	0.0008** (0.0004)	0.0002 (0.0010)	0.0009 (0.0007)	-0.0009 (0.0009)	0.0007** (0.0003)
Undergraduate GPA	0.1609*** (0.0551)	0.0264 (0.0208)	0.1022** (0.0423)	0.0484 (0.0522)	0.0757* (0.0418)	0.0488*** (0.0176)
TOEFL/IELTS	0.0297 (0.0219)	0.0244 (0.0179)	0.0821** (0.0417)	0.0523* (0.0301)	-0.0198 (0.0259)	0.0326*** (0.0123)
Elite U.S. University	0.0677*** (0.0225)	0.0149 (0.0119)	0.0490* (0.0281)	-0.0328 (0.0264)	0.0630** (0.0271)	0.0233** (0.0090)
Elite U.S. Liberal Arts College	0.0855*** (0.0288)	0.0238* (0.0143)	0.0801** (0.0402)	-0.0330 (0.0761)	0.0987** (0.0480)	0.0429*** (0.0136)
Elite Foreign University	0.0770* (0.0463)	0.0180 (0.0311)	0.1327** (0.0591)	0.0877* (0.0481)	0.0166 (0.0850)	0.0479** (0.0241)
Other Foreign University	0.0416 (0.0441)	0.0219 (0.0201)	0.1026* (0.0527)	0.0227 (0.0391)	0.0458 (0.0556)	0.0273 (0.0221)
Graduate Degree	0.0036 (0.0171)	0.0028 (0.0098)	0.0234 (0.0247)	0.0219 (0.0262)	0.0208 (0.0273)	0.0106 (0.0089)
Work Experience	0.0301** (0.0144)	0.0152 (0.0113)	0.0372 (0.0246)	0.0061 (0.0205)	0.0776** (0.0368)	0.0221** (0.0099)
#Professor Recommenders	0.0026 (0.0101)	-0.0007 (0.0051)	0.0413** (0.0190)	0.0259* (0.0142)	-0.0042 (0.0113)	0.0063 (0.0047)
#Profic Recommenders	0.0202*** (0.0065)	0.0114** (0.0048)	0.0141 (0.0118)	0.0178** (0.0082)	0.0251* (0.0130)	0.0133*** (0.0038)
#Math Courses	0.0129*** (0.0045)					
Advanced Math	0.0177 (0.0137)					
Missing GRE	0.0145 (0.0585)	-0.0123 (0.0118)	0.0470 (0.0319)	-0.0045 (0.0204)	-0.0491* (0.0252)	-0.0062 (0.0088)
Missing Undergraduate GPA	-0.0363 (0.0375)	-0.0186 (0.0206)	-0.1169*** (0.0524)	-0.0601** (0.0355)	0.0200 (0.0505)	-0.0291 (0.0179)
Missing TOEFL/IELTS	-0.0290 (0.0188)	0.0234 (0.0215)	0.0505 (0.0388)	-0.0024 (0.0276)	-0.0734* (0.0417)	0.0108 (0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	115.8837	23	0.0000			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

A4: Graduate School Logistic Regression Results (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997 (0.1633)	-0.0323 (0.2215)	0.2028 (0.3078)	-0.4791 (0.3477)	-0.5946 (0.4520)
Winsorized Age	-0.0509** (0.0201)	-0.0198 (0.0174)	0.0576* (0.0307)	-0.0164 (0.0357)	0.1203*** (0.0400)
Female	0.1941 (0.1402)	0.5023 (0.2004)	0.1389 (0.2526)	0.2595 (0.3255)	0.2119 (0.4310)
U.S. African/Hispanic/Native	0.3053 (0.3724)	-0.1457 (0.4303)	-0.2379 (0.4670)	-1.6401 (1.0720)	0.5413 (0.5306)
U.S. Asian	0.5663 (0.3642)	0.6016 (0.4452)	-0.7934 (1.1675)	-0.4305 (0.7939)	0.7122 (0.8863)
International Asian	-0.1428 (0.2169)	0.1630 (0.2803)	0.5780 (0.3740)	-0.4965 (0.4275)	0.1866 (0.6866)
International Other	-0.1231 (0.2311)	0.1395 (0.3046)	0.1097 (0.3237)	-0.1602 (0.3832)	-1.0807 (0.7583)
GRE Verbal Percentile	0.0167*** (0.0046)	0.0247** (0.0117)	0.0164 (0.0127)	-0.0029 (0.0100)	0.0132 (0.0184)
GRE Quantitative Percentile	0.0233*** (0.0090)	-0.0027 (0.0051)	0.0026 (0.0088)	0.0304** (0.0137)	0.0097 (0.0099)
GRE Analytic Percentile	0.0042 (0.0031)	0.0004 (0.0059)	-0.0073 (0.0062)	-0.0042 (0.0101)	0.0069 (0.0103)
Undergraduate GPA	0.2292 (0.4745)	0.6327 (0.4085)	1.3552** (0.5706)	-1.3171** (0.5584)	-0.1802 (0.6152)
TOEFL/IELTS	-0.4673** (0.2157)	0.5141 (0.4185)	-0.7648 (0.6814)	0.2446 (0.7498)	-0.7354 (0.8025)
Elite U.S. University	-0.4850* (0.2560)	-0.2243 (0.3258)	0.0572 (0.3421)	0.4764 (0.4788)	0.3941 (0.4762)
Elite U.S. Liberal Arts College	0.2429 (0.3049)	0.1759 (0.4042)	0.2358 (0.4964)	0.4860 (0.9631)	0.8652 (0.7261)
Elite Foreign University	-0.1622 (0.3015)	-1.1920 (0.7484)	0.7404 (0.5668)	0.4801 (0.7841)	3.0530*** (0.8673)
Other Foreign University	0.1071 (0.1979)	0.2042 (0.2732)	0.2721 (0.3301)	0.1731 (0.4202)	0.1731 (0.5926)
Graduate Degree	-0.0198 (0.1624)	0.1770 (0.2439)	-0.1601 (0.2765)	-0.8608*** (0.3320)	-0.2769 (0.3814)
Work Experience	0.2525* (0.1441)	-0.0391 (0.2396)	-0.0736 (0.2629)	-0.2650 (0.3417)	-1.1936** (0.4805)
@Professor Recommenders	-0.1059 (0.0909)	0.0011 (0.1138)	0.1828 (0.1473)	0.2181 (0.1817)	-0.4923** (0.1956)
@Prolific Recommenders	0.0039 (0.0956)	0.2189** (0.1108)	-0.1743 (0.1723)	0.0673 (0.1642)	0.5134** (0.2213)
@Math Courses	0.0366 (0.0421)				
Advanced Math	0.1961 (0.1416)				
Missing GRE	-0.1993 (0.4328)	-0.2020 (0.2195)	-0.0011 (0.2689)	-0.0433 (0.3503)	-0.6040 (0.4817)
Missing Undergraduate GPA	0.1531 (0.1548)	0.0895 (0.2291)	-0.2455 (0.3015)	0.6298** (0.3125)	-0.3574 (0.4342)
Missing TOEFL/IELTS	-0.2950* (0.1640)	0.4898 (0.3480)	0.5752 (0.5466)	0.1346 (0.4345)	-2.0781*** (0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.0524	0.0396	0.0753	0.1046	0.2230

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive** ($p < .1$): GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative** ($p < .1$): Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A4: Graduate School Logistic Regression AMEs (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204 (0.0167)	-0.0014 (0.0099)	0.0192 (0.0291)	-0.0298 (0.0216)	-0.0345 (0.0262)	-0.0089 (0.0086)
Winsorized Age	-0.0052** (0.0021)	-0.0009 (0.0008)	0.0054* (0.0029)	-0.0010 (0.0022)	0.0070*** (0.0023)	0.0004 (0.0008)
Female	0.0108 (0.0143)	0.0041 (0.0239)	0.0131 (0.0294)	0.0162 (0.0251)	0.0123 (0.0078)	0.0099 (0.0078)
U.S. African/Hispanic/Native	0.0312 (0.0381)	-0.0065 (0.0192)	-0.0225 (0.0441)	-0.1021 (0.0673)	0.0314 (0.0306)	-0.0137 (0.0147)
U.S. Asian	0.0579 (0.0372)	0.0268 (0.0199)	-0.0750 (0.1101)	-0.0268 (0.0495)	0.0413 (0.0516)	0.0241 (0.0179)
International Asian	-0.0146 (0.0222)	0.0073 (0.0125)	0.0546 (0.0353)	-0.0309 (0.0263)	0.0108 (0.0399)	0.0075 (0.0112)
International Other	-0.0126 (0.0236)	0.0062 (0.0136)	0.0104 (0.0306)	-0.0100 (0.0237)	-0.0627 (0.0443)	-0.0014 (0.0106)
GRE Verbal Percentile	0.0017*** (0.0005)	0.0011** (0.0005)	0.0015 (0.0012)	-0.0002 (0.0006)	0.0008 (0.0010)	0.0008** (0.0004)
GRE Quantitative Percentile	0.0024*** (0.0009)	-0.0001 (0.0002)	0.0003 (0.0008)	0.0019*** (0.0009)	0.0006 (0.0006)	0.0003 (0.0002)
GRE Analytic Percentile	0.0004 (0.0003)	0.0000 (0.0003)	-0.0007 (0.0009)	-0.0003 (0.0006)	0.0004 (0.0006)	0.0000 (0.0002)
Undergraduate GPA	0.0234 (0.0485)	0.0282 (0.0223)	0.1281** (0.0553)	-0.0020** (0.0358)	-0.0058 (0.0357)	0.0130 (0.0158)
TOEFL/IELTS	-0.0478** (0.0221)	0.0229 (0.0187)	-0.0723 (0.0645)	0.0152 (0.0466)	-0.0427 (0.0469)	0.0050 (0.0187)
Elite U.S. University	-0.0496* (0.0261)	-0.0100 (0.0145)	0.0054 (0.0244)	0.0297 (0.0297)	0.0229 (0.0274)	0.0036 (0.0108)
Elite U.S. Liberal Arts College	0.0248 (0.0312)	0.0078 (0.0180)	0.0223 (0.0469)	0.0303 (0.0602)	0.0502 (0.0418)	0.0217 (0.0160)
Elite Foreign University	-0.0166 (0.0308)	-0.0532 (0.0337)	0.0700 (0.0537)	0.0299 (0.0490)	0.1771*** (0.0484)	0.0158 (0.0179)
Other Foreign University	-0.0111 (0.0202)	0.0091 (0.0122)	0.0018 (0.0321)	0.0015 (0.0267)	0.0175 (0.0344)	0.0116 (0.0103)
Graduate Degree	-0.0020 (0.0166)	0.0079 (0.0109)	-0.0151 (0.0261)	-0.0542*** (0.0206)	-0.0161 (0.0220)	-0.0093 (0.0085)
Work Experience	0.0258* (0.0148)	-0.0017 (0.0107)	-0.0070 (0.0248)	-0.0165 (0.0171)	-0.0692** (0.0275)	-0.0093 (0.0086)
#Professor Recommenders	-0.0108 (0.0093)	0.0041 (0.0051)	-0.0173 (0.0140)	0.0136 (0.0114)	-0.0286*** (0.0110)	-0.0027 (0.0042)
#Profic Recommenders	0.0004 (0.0067)	0.0008* (0.0050)	-0.0165 (0.0164)	0.0042 (0.0102)	0.0298** (0.0128)	0.0083* (0.0045)
#Math Courses	0.0237 (0.0043)					
Advanced Math	0.0201 (0.0145)					
Missing GRE	-0.0204 (0.0443)	-0.0090 (0.0086)	-0.0001 (0.0254)	-0.0027 (0.0218)	-0.0390 (0.0239)	-0.0098 (0.0079)
Missing Undergraduate GPA	0.0157 (0.0158)	0.0040 (0.0102)	0.0392* (0.0285)	-0.0207 (0.0201)	-0.0207 (0.0254)	0.0019 (0.0083)
Missing TOEFL/IELTS	-0.0302* (0.0168)	0.0218 (0.0156)	0.0544 (0.0517)	0.0084 (0.0272)	-0.1205*** (0.0434)	0.0063 (0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	37.4658	23	0.0290			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A4: Graduate School Lasso Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Winsorized Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R ²	0.2297	0.0912	0.0592	0.0850	0.0537

- **Economics Positive:** Female, U.S. African/Hispanic/Native, International Other, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, TOEFL/IELTS, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, # Prolific Recommenders, #Math Courses, Advanced Math
- **Economics Negative:** Covid Year, Winsorized Age, U.S. Asian, International Asian, Missing TOEFL/IELTS

References I



Anderson, A., J. Rosen, J. Rust & K.-P. Wong (2025). "Disequilibrium Play in Tennis". In: *Journal of Political Economy* 133.1, pp. 190–251.



Athey, S., L. F. Katz, A. B. Krueger, S. Levitt & J. Poterba (2007). "What Does Performance in Graduate School Predict? Graduate Economics Education and Student Outcomes". In: *American Economic Review* 97.2, pp. 512–520.



Athey, S. & M. Luca (2019). "Economists (and Economics) in Tech Companies". In: *Journal of Economic Perspectives* 33.1, pp. 209–230.



Bai, J., M. Esche, W. B. MacLeod & Y. Shi (2022). "Subjective Evaluations and Stratification in Graduate Education". In: *NBER Working Paper* 30677.



Caplin, A. & M. Dean (2015). "Revealed Preference, Rational Inattention, and Costly Information Acquisition". In: *American Economic Review* 105.7, pp. 2183–2203.



Cho, S. & J. Rust (2010). "The Flat Rental Puzzle". In: *Review of Economic Studies* 77.2, pp. 560–594.



Heckman, J. J. (1979). "Sample Selection Bias as a Specification Error". In: *Econometrica* 47.1, pp. 153–161.



Hoffman, M., L. B. Kahn & D. Li (2018). "Discretion in Hiring". In: *The Quarterly Journal of Economics* 133.2, pp. 765–800.



Kleinberg, J., H. Lakkaraju, J. Leskovec, J. Ludwig & S. Mullainathan (2018). "Human Decisions and Machine Predictions". In: *The Quarterly Journal of Economics* 133.1, pp. 237–293.



Krueger, A. B. & S. Wu (2000). "Forecasting Job Placements of Economics Graduate Students". In: *The Journal of Economic Education* 31.1, pp. 81–94.



Lewis, M. (2004). *Moneyball: The Art of Winning an Unfair Game*. W.W. Norton & Company.



Madera, J. M., M. R. Hebl & R. C. Martin (2009). "Gender and Letters of Recommendation for Academia: Agentic and Communal Differences". In: *Journal of Applied Psychology* 94.6, pp. 1591–1599.

References II



Matějka, F. & A. McKay (2015). "Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model". In: *American Economic Review* 105.1, pp. 272–298.



McCall, J. (1970). "Economics of Information and Job Search". In: *The Quarterly Journal of Economics* 84.1, pp. 113–126.



Mood, C. (2010). "Logistic Regression: Why We Cannot Do What We Think We Can Do, and What We Can Do About It". In: *European Sociological Review* 26.1, pp. 67–82.



Mortensen, D. T. (1970). "Job Search, the Duration of Unemployment, and the Phillips Curve". In: *American Economic Review* 60.5, pp. 847–862.



Mullainathan, S. & J. Spiess (2017). "Machine Learning: An Applied Econometric Approach". In: *Journal of Economic Perspectives* 31.2, pp. 87–106.



Patki, N., R. Wedge & K. Veeramachaneni (2016). "The Synthetic data vault". In: *IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 399–410.



Sims, C. A. (2003). "Implications of rational inattention". In: *Journal of Monetary Economics* 50.3, pp. 665–690.



Stock, W. A., T. A. Finegan & J. J. Siegfried (2006). "Attrition in Economics Ph.D. Programs". In: *American Economic Review* 96.2, pp. 458–466.



Stock, W. A. & J. J. Siegfried (2015). "The Undergraduate Origins of PhD Economists Revisited". In: *The Journal of Economic Education* 46.2, pp. 150–165.



Weitzman, M. L. (1979). "Optimal Search for the Best Alternative". In: *Econometrica* 47.3, pp. 641–654.



Young, L. & S. Soroka (2012). "Affective News: The Automated Coding of Sentiment in Political Texts". In: *Political Communication* 29.2, pp. 205–231.